

Parking Problems—Solutions

Spring 2019 ARML Power Contest

Problem 1. (a) Each of the two drivers can prefer space 1 or space 2, so there are four possibilities: $(1, 1)$, $(1, 2)$, $(2, 1)$, and $(2, 2)$.

(b) In each of the first three cases, both drivers can park. For $(2, 2)$ both drivers bypass space 1, but can't park in the same space, so this is not a parking sequence.

Problem 2. (a) The first four drivers get their choice of space, but the last two cannot park, so the spaces are filled in order $\boxed{4563xx}$.

(b) The spaces fill up in the order 2, 4, 3, 5, 6 but the last driver can't park. The result: $\boxed{24356x}$.

(c) This time, all drivers can park. The order the spaces fill is $\boxed{342516}$.

Problem 3. Since each of n drivers can prefer any of the n spaces, the multiplication principle tells us that the number of preference sequences is n^n .

Problem 4. Tediously, we list out all the possibilities.

$(1, 1, 1)$ $(1, 1, 2)$ $(1, 2, 1)$ $(2, 1, 1)$ $(1, 1, 3)$ $(1, 3, 1)$ $(3, 1, 1)$ $(1, 2, 2)$
 $(2, 1, 2)$ $(2, 2, 1)$ $(1, 2, 3)$ $(1, 3, 2)$ $(2, 1, 3)$ $(2, 3, 1)$ $(3, 1, 2)$ $(3, 2, 1)$

Problem 5. We will use brute force to consider all the possibilities. Let $p_j = x$ and $p_{j+1} = y$ be the places where the drivers prefer to park. So we need to consider the effect of changing preference sequence $(\dots, x, y, \dots)$ to $(\dots, y, x, \dots)$. If $x = y$ so that the two drivers who swapped positions wanted the same space, it is clear the preference sequence does not change, so whether or not it is a parking sequence can't change either.

Assume the original sequence is a parking sequence and driver j ends up parking in space s while $j + 1$ ends up in space t . Note $x \leq s$ and $y \leq t$.

Case 1: $x \leq s < y$. Then, after the change in driver order, driver $j + 1$ has exactly the same choices as before. Because even though they now drive in before driver j , the space, s , that is now empty is before y and thus $j + 1$ won't park there. So $j + 1$ parks in space t as before. This leaves spaces s still available to driver j , and no changes result, so the new order is still a parking sequence.

Case 2: $x < y \leq s$. Note that $t > s$ because otherwise j would have parked in space t in the original driver order. So now the earliest space available to driver $j + 1$ is space s . Now space s isn't available to driver j —just like it wasn't available to $j + 1$ in the original order since j had just parked there. But now j can proceed to park in space t , since that is now the earliest available space. The rest of the cars now park as in the original order, and once again the result is a parking sequence.

Case 3: $y \leq t < x$. Once again, all drivers park in the same spots as originally; driver j sacrifices nothing by allowing $j + 1$ go first, because the space that driver $j + 1$ parks in is lower-numbered than j wants.

Case 4: $y < x \leq t$. Note that, as in case 2 above, $s < t$ or driver j would have originally parked in t and not s . So we actually have $y < x \leq s < t$. In this case, the first space available to $j + 1$ now that they are going first is space s . But, again as in case 2, j will end up parking in t and no other driver's spaces is affected, and the resulting sequence is again a parking sequence.

These are all the possibilities when the original sequence was a parking sequence, and in each case the resulting sequence was again a parking sequence. So a swap of two drivers cannot change a parking sequence into a non-parking sequence. Nor can it change a non-parking sequence into a parking sequence, or swapping the drivers back to their original positions would mean that the original sequence was also a parking sequence. So a swap of driver positions does not affect whether or not a sequence is a parking sequence.

Problem 6. We will prove this in the case $a = 1$ and then apply the result recursively for $a > 1$. That is, we will prove the result in case one driver decreases their preference by one.

First, if space $p_j - 1$ is already occupied, the change in preference has no effect on the outcome. Driver j must drive past the spot and park in p_j or later, just as if their preference had not changed.

On the other hand, if $p_j - 1$ is available, that must mean in the original preference list there is some driver k with $k > j$ who parked in that space. For every driver k with $k > j$ who, in the original preference list, parked in a space earlier than p_j , use the previous problem to let that driver come before driver j . This will cause all the spaces from 1 to $p_j - 1$ to become full before driver j gets to park, and so space $p_j - 1$ isn't available. The previous paragraph then assures us we still have a parking sequence.

Problem 7. First, note that the preference sequence $(1, 2, \dots, n)$ is a parking sequence. By the previous problem, any of these numbers can be reduced and leave a parking sequence. So if a sequence can be sorted into one where each term is less than or equal to the corresponding term of $(1, 2, \dots, n)$ it must be a parking sequence.

On the other hand, if (when sorted) any $p_j > j$ then there are at least $p - (j - 1) = p - j + 1$ cars that must park in the $p - j$ spots numbered $p - j + 1$ and higher. By the pigeonhole principle some car will not be able to park.

An alternate proof of the "if" portion can be obtained by induction. For $k < n$ assume that the first k cars have parked in spaces 1 through k . Driver $k + 1$ has preference for a space numbered at most $k + 1$. Spaces 1 through k are already full, so driver $k + 1$ will park in space $k + 1$. (This is a strong induction that doesn't actually need a base case (the statement is vacuously true), though the case $k = 1$ is clearly true if the vacuous base case makes you nervous.) So the n cars will park in spaces 1 through n , and the sequence is a parking sequence.

Problem 8. The spaces will fill in order 2, 4, 3, 5, 6, 7.

Problem 9. There is more than one possible answer for each k ; these are samples. For $k = 1$: (2, 3, 4, 5, 5). For $k = 2$: (1, 3, 4, 5, 5). For $k = 3$: (1, 2, 4, 5, 5). For $k = 4$: (1, 2, 3, 5, 5).

Problem 10. The only way a car could park in space $n + 1$ is it had an earlier preference, say p , but spaces p through n were already full. This implies that to have car in space $n + 1$, space n must be full. But since there are n cars and $n + 1$ spaces, it is not possible for both spaces n and $n + 1$ to be empty by the pigeonhole principle.

Problem 11. The argument is exactly the same as the previous problem. Since space k is empty, space $k + 1$ must be occupied. There are only two ways for this to happen: a driver prefers that space, or a driver had to bypass space k because it was full. The latter is a contradiction, so it must be the former.

Problem 12. For preference list $(p_1, p_2, \dots, p_n)$ let $[s_1, s_2, \dots, s_n]$ be the list of the spaces in the order in which a car parks in them. We will show by induction that $(p_1 + a, p_2 + a, \dots, p_n + a)$ leads to $[t_1, t_2, \dots, t_n]$ with $t_j = s_j + a$. Clearly, since there are no cars parked yet when the first driver comes, they can park in their preferred spot. So $s_1 = p_1$ and $t_1 = p_1 + a = s_1 + a$, so the base case is established.

Now assume that for some $r < n$ we have $t_1 = s_1 + a, t_2 = s_2 + a, \dots, t_r = s_r + a$. Then either spot p_{r+1} is unoccupied in the original garage—in which case $p_{r+1} + a$ is unoccupied in the shifted garage, and $t_{r+1} = p_{r+1} + a = s_{r+1} + a$. Otherwise, by induction, the occupied spaces in the shifted garage are the shifts of the occupied spaces in the original garage, and the first open space available for driver $r + 1$ will be the shift by a of the first open space in the original garage. That is, $t_{r+1} = s_{r+1} + a$ in this case also, completing the induction.

Since the occupied spaces are all shifted by a , the unoccupied space must also be shifted by a , as required by the problem.

Problem 13.

(a) Since space $n + 1$ is empty, no driver can prefer that space, so this garage preference sequence only contains numbers from 1 to n , and is thus a standard preference sequence.

(b) If the sequence is not a parking sequence, some driver would not be able to park in any of the spaces 1 through n . Since their preference is at most n , the first available space would be $n + 1$. So if the sequence is not a parking sequence, space $n + 1$ will be occupied.

Problem 14. Assemble the previous questions into a complete explanation. Given any garage preference sequence $(p_1, p_2, \dots, p_n)$, each shift $(p_1 + a, \dots, p_n + a)$ leaves a different space empty (problem 12). So exactly one of the $n + 1$ possible shifts (including a shift by zero) leaves space $n + 1$ unoccupied. By problem 13 this shift must be a parking sequence. Now there are $(n + 1)^n$ possible garage preference sequences (each of the n drivers has $n + 1$ spaces they can select as their preference). Exactly $1/(n + 1)$ of these is a parking sequence (and every parking sequence is a garage preference sequence, so every parking sequences arises in this way), leaving a total of $(n + 1)^{n-1}$ parking sequences.

Problem 15. The sequence $(1, 2, \dots, n)$ is a parking sequence, and by problem 5 any of the $n!$ re-orderings of this sequence is also a parking sequence. But for $n \geq 2$, there are other parking sequences, such as $(1, 1, \dots, 1)$. So there are more parking sequences than arrangements of $(1, 2, \dots, n)$, hence $(n+1)^{n-1} > n!$.

Problem 16. The three non-parallel parking sequences are $(3, 3, 3)$, $(2, 3, 3)$, and $(3, 2, 3)$.

Problem 17. From the previous problem we know that $(2, 3, 3)$ is not a parallel parking sequence, but its rearrangement $(3, 3, 2)$ is: the first driver parks in space 3, the second tries to park there but has to back into space 2, and then the last driver tries to park in space 2 but has to back into space 1.

Problem 18. If driver j initially parked in a space numbered $p_{j+1}-2$ or less, then driver $j+1$ cannot park in that space even if they enter the lot before driver j and end up backing into a space, so exactly the same parking spaces will be occupied by the same cars regardless of the order of drivers j and $j+1$. On the other hand, if driver j initially ends up in space $p_{j+1}-1$ or higher, then driver $j+1$ will get that space first, and then driver j will get the spaces that would have originally gone to $j+1$. (Exception: if spaces p_{j+1} and $p_{j+1}-1$ are both available, driver $j+1$ will get space p_{j+1} and driver j will get $p_{j+1}-1$ regardless of the order in which they enter the street.) In any case, all drivers will be able to park, so the sequence remains a parallel parking sequence.

Problem 19. Consider a preference sequence that is not a parallel parking sequence. Let driver i be the first driver who is unable to park. Then spaces p_i-1 through n (a total of $n-p_i+2$) are all already taken. Since a driver can back up at most one space when parking, if the ability to back up were taken away from the drivers, each car in one of these spaces would have to have parked in one of the spaces p_i through n under the original parking rules. But there are $n-p_i+1$ such spaces for $n-p_i+2$ cars, so by the pigeonhole principle someone doesn't get to park under the original parking rules. Thus any non-parallel parking sequence is not a parking sequence, and conversely each parking sequence must be a parallel parking sequence.

Problem 20. There are 203 such sequences. This can be arrived at as follows.

- Any sequence of length 4 having at least two numbers less than or equal to 2 is a parallel parking sequence. For the first two such drivers will get spaces 1 and 2, and then the other two drivers can fit into spaces 3 and 4. This is clear from the fact that all four sequences of length 2 are parallel parking sequences.
 - There are $2^4 = 16$ sequences where we choose each of the four entries to be either 1 or 2.
 - There are 32 sequences with a single 3 (four places to put the 3 and $2^3 = 8$ choices to make the other three numbers either 1 or 2).
 - There are another 32 sequences with a single 4.
 - For sequences with two 3's or 4's, there are $\binom{4}{2} = 6$ choices of where to put them, times $2^2 = 4$ choices of whether each preference is a 3 or a 4, and times $2^2 = 4$ choices of the two small numbers being 1 or 2. That leads to $6 \cdot 4 \cdot 4 = 96$ possibilities.

So altogether this gives 176 sequences.

- $(3, 3, 3, 1)$, $(3, 3, 1, 3)$, $(3, 1, 3, 3)$, $(1, 3, 3, 3)$, and $(3, 3, 3, 2)$, for five more.
- Each sequence above can have any one of the 3's replaced by a 4. Also, both of the first two 3's can be replaced by 4's. That adds another 20 sequences.
- The two remaining parallel parking sequences are $(3, 3, 2, 3)$ and $(3, 3, 2, 4)$.

There are several combinatorial problems whose answer is $(n+1)^{n-1}$. A couple interesting ones are:

- The number of spanning trees of a complete graph on n distinct point.
- If you take items numbered from 1 to n and place them randomly in a circle, this is the number of ways you could make exactly $n - 1$ swaps of positions of two objects in the circle, and end up with the circle sorted from 1 to n in a clockwise order.
- If you take a sequence of n integers from 1 to n , this is the number of such sequences where the first few terms add up to n . That is, of the 27 sequences of three numbers from 1 to 3 (i.e., “preference sequences” in the terminology of this problem), there are 16 of them where the first few terms add to 3: $(1, 1, 1)$, $(1, 2, x)$, $(2, 1, x)$ and all nine sequences of the form $(3, x, y)$.

Incredibly, the parking sequences are also closely related to “critical arrangements” of the chip firing game from the fall contest! These are configurations that are stable—no pile can fire—but with the addition of a single chip the configuration is able to fire and never becomes stable after any finite number of firings.